

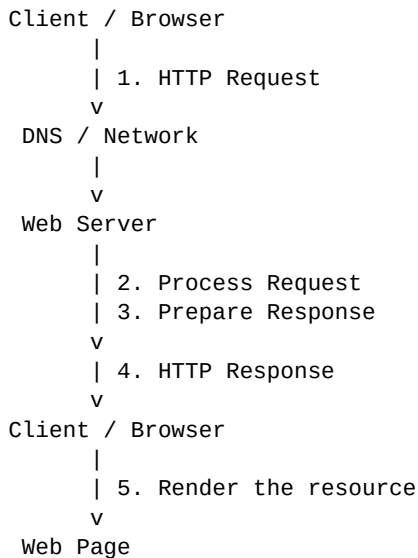
Frontend Assignment Theory 1

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Q1. HTTP Request–Response Cycle

Answer: The HTTP Request–Response Cycle is the process through which a client such as a web browser communicates with a web server to obtain a resource such as an HTML page, CSS file, image, or JSON data.



Major steps: (1) The user enters a URL or clicks a link. (2) The browser determines the server address, commonly using DNS. (3) The browser establishes the required network connection and sends an HTTP request containing information such as the method, URL, headers, and sometimes a body. (4) The web server receives and processes the request. (5) The server sends an HTTP response containing a status code, headers, and usually a response body. (6) The browser receives the response and renders the page, making additional requests for resources such as CSS, JavaScript, and images when necessary.

Q2. Introduction to Web Technologies

1. HTML: HTML (HyperText Markup Language) provides the structure and content of a web page. It uses elements such as headings, paragraphs, links, images, forms, and tables.

2. CSS: CSS (Cascading Style Sheets) controls the presentation of HTML. It is used for colors, fonts, spacing, borders, layouts, animations, and responsive design.

3. JavaScript: JavaScript is a programming language used to add behavior and interactivity to web pages. It can respond to events, modify the DOM, validate forms, and communicate with servers.

4. Bootstrap: Bootstrap is a frontend framework that provides ready-made CSS components, a responsive grid system, utility classes, and JavaScript-powered components for building responsive interfaces quickly.

5. jQuery: jQuery is a JavaScript library that simplifies common tasks such as DOM manipulation, event handling, animations, and AJAX requests.

6. React.js: React.js is a JavaScript library for building user interfaces using reusable components. It is particularly useful for developing dynamic, component-based frontend applications.

Q3. Client-Side vs. Server-Side Technologies

Client-side technologies run mainly in the user's browser. They are responsible for the interface and interactions visible to the user. Examples include HTML, CSS, and JavaScript.

Server-side technologies run on the web server. They handle tasks such as business logic, authentication, database operations, and generating or supplying data. Examples include Node.js, Java (Servlets/Spring), Python (Django/Flask), PHP, and .NET.

Feature	Client-Side	Server-Side
Runs on	User's browser	Web server
Main purpose	UI, presentation and interaction	Business logic, data processing and database access
Examples	HTML, CSS, JavaScript	Node.js, Java, Python, PHP
Visibility	Code may be sent to the browser	Implementation normally remains on the server

Q4. Block-Level Elements in HTML

Answer: Block-level elements normally begin on a new line and occupy the available width of their containing block by default. They are commonly used to organize the major structure of a web page. CSS can change their display behavior.

Characteristics: They normally start on a new line, can contain other elements depending on HTML rules, and are suitable for creating page sections and structural layouts.

Examples: <div>, <p>, <h1>, <section>, <header>, and <footer>.

Q5. Semantic HTML Elements

Answer: Semantic elements clearly describe the meaning or purpose of their content. They make the structure of a page easier for developers, browsers, search engines, and assistive technologies to understand.

Importance: Semantic HTML improves code readability and maintainability, provides a meaningful document structure, supports accessibility, and can help search engines understand page content.

Examples: <header>, <nav>, <main>, <section>, <article>, <aside>, and <footer>.

Q6. Creating a Table Using HTML

Answer: The following HTML creates the required table with headings, rows, and columns.

```
<!DOCTYPE html>
<html>
<head>
  <title>Employee Table</title>
</head>
<body>
  <table border="1">
    <tr>
      <th>Sr. No.</th>
      <th>Name</th>
      <th>Designation</th>
      <th>Department</th>
    </tr>
    <tr>
      <td>1</td>
      <td>John</td>
```

```

        <td>Manager</td>
        <td>Sales</td>
    </tr>
    <tr>
        <td>2</td>
        <td>Smith</td>
        <td>Sr. Manager</td>
        <td>Marketing</td>
    </tr>
    <tr>
        <td>3</td>
        <td>Jane</td>
        <td>Manager</td>
        <td>HR</td>
    </tr>
</table>
</body>
</html>

```

Q7. Creating a Nested List Using HTML

Answer: An unordered list can be nested inside an ordered or unordered list. For the requested structure, the following code can be used:

```

<ul>
    <li>River
        <ul>
            <li>Ganga</li>
            <li>Yamuna</li>
            <li>Brahmaputra</li>
            <li>Narmada</li>
        </ul>
    </li>
</ul>

```

Q8. Types of CSS

1. Inline CSS: CSS is written directly inside an HTML element using the style attribute. It is useful for applying a style to one specific element.

```
<p style="color: blue;">Hello World</p>
```

2. Internal CSS: CSS is written inside a <style> element in the HTML document's <head>. It is useful when styles are intended for one page.

```

<style>
    p {
        color: blue;
        font-size: 18px;
    }
</style>

```

3. External CSS: CSS is stored in a separate .css file and connected to HTML using a <link> element. It is preferred for larger websites because the same stylesheet can be reused across multiple pages.

```

<link rel="stylesheet" href="style.css">

/* style.css */
p {
    color: blue;
}

```

Q9. Practical Understanding

HTML: HTML provides the structure and content of the web page. It defines elements such as headings, paragraphs, images, links, tables, and forms.

CSS: CSS provides the visual presentation. It controls layout, colors, typography, spacing, sizing, responsiveness, and other visual effects.

JavaScript: JavaScript provides behavior and interactivity. It can handle user actions, change page content dynamically, validate input, perform calculations, and communicate with web services.

Technology	Primary Role	Example Contribution
HTML	Structure	Creates headings, paragraphs, forms and tables
CSS	Presentation	Controls colors, layout, fonts and responsive design
JavaScript	Behavior	Adds events, dynamic updates and application logic

Q10. Short Answer

a. What is the role of Bootstrap in web development?

Bootstrap is a frontend framework that helps developers build responsive and consistent user interfaces quickly. It provides a grid system, reusable components, utility classes, and responsive design features.

b. What is jQuery and why was it widely used in web development?

jQuery is a JavaScript library that provides a simpler API for tasks such as selecting and modifying HTML elements, handling events, creating animations, and making AJAX requests. It became widely used because it reduced the amount of code required for common frontend tasks and helped handle browser differences.

c. What is React.js and what problem does it solve in frontend development?

React.js is a JavaScript library for building user interfaces from reusable components. It helps developers manage complex, dynamic interfaces by organizing the UI into components and efficiently updating the parts of the page that need to change.